



AI & Quantum Computing: Emerging Paradigms in Biomedical Research & Personalized Medicine

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Abstract

Biomedical research is witnessing a paradigm shift with the advent of AI and quantum computing. AI driven algorithms, machine learning, and deep learning are adept at analysing huge amount of unstructured data generated in labs and hospitals regularly. These can efficiently identify disease patterns, mechanisms, biomarkers and therapeutic targets aiding in diagnosis and prognosis. Quantum computing, leveraging quantum mechanics, and quantum algorithms accelerates molecular simulations and drug discovery by simulating molecular interactions and predicting drug efficacy. The integration of AI and quantum computing fosters a future of precise and patient-centric healthcare. The synergy between AI's capabilities of analysing patient's specific data and Quantum's computational power to design personalised drug therapies tailored to match specific genetic variations promises to revolutionize personalized medicine and healthcare. This approach enhances biomedical research by complementing the diagnostic accuracy and efficient therapeutics development.

Introduction

Artificial Intelligence

Artificial Intelligence (AI) is the ability of a computer program or a machine to think and learn similar, to the way that human does naturally and be as cognitive as possible via its Artificial Neural Network (ANN) system built in the machine. [1]

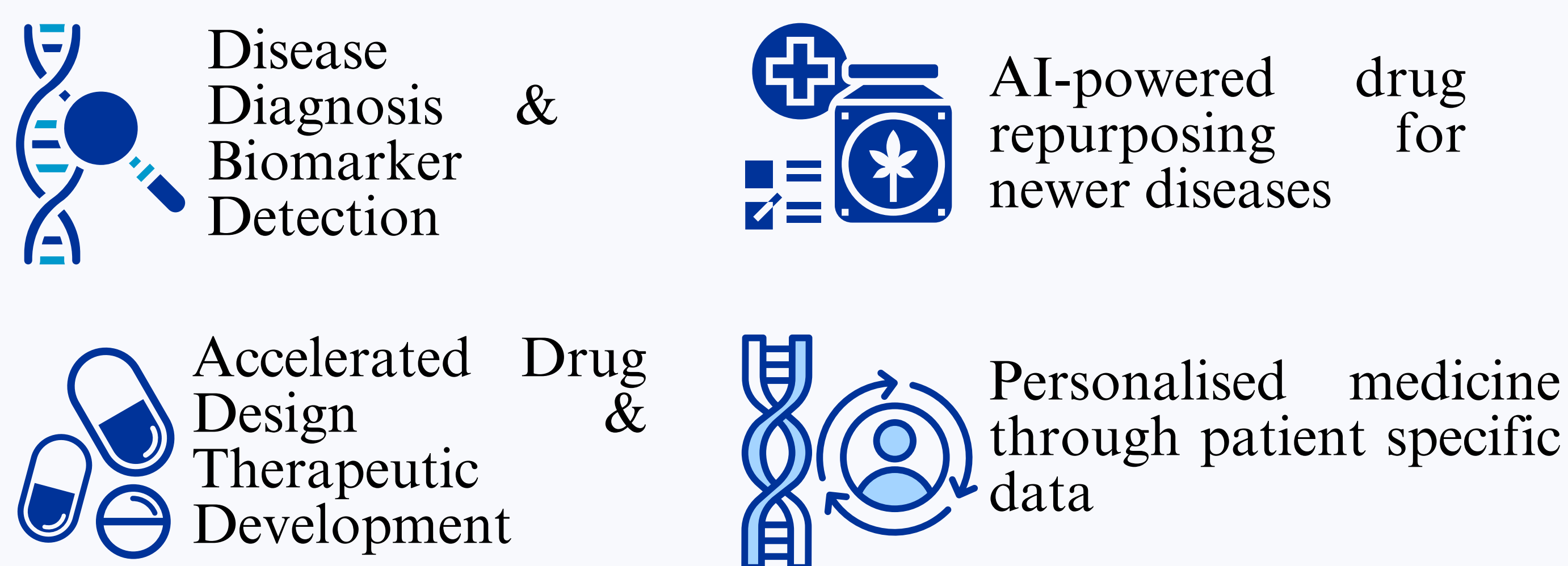
Quantum Computing

Quantum computing (QC) is a concept in computer science that utilizes the principles of quantum physics to perform computations at a much faster rate than traditional computers.[2]

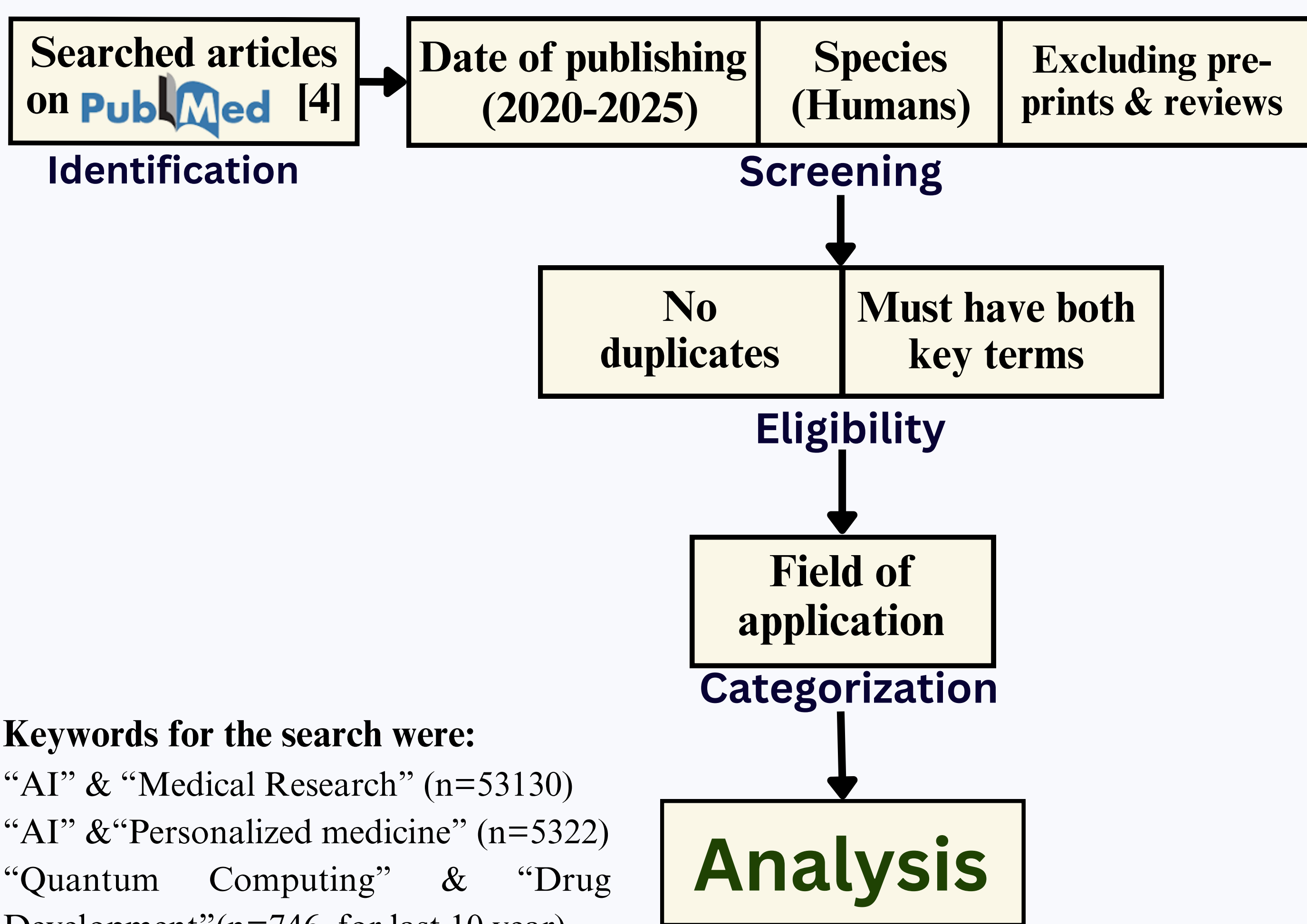
Personalized Medicine

Precision medicine, sometimes known as "personalized medicine" is an innovative approach to tailoring disease prevention and treatment that takes into account differences in people's genes, environments, and lifestyles.[3]

Objectives



Methodology



Application of AI in Research & Therapeutics

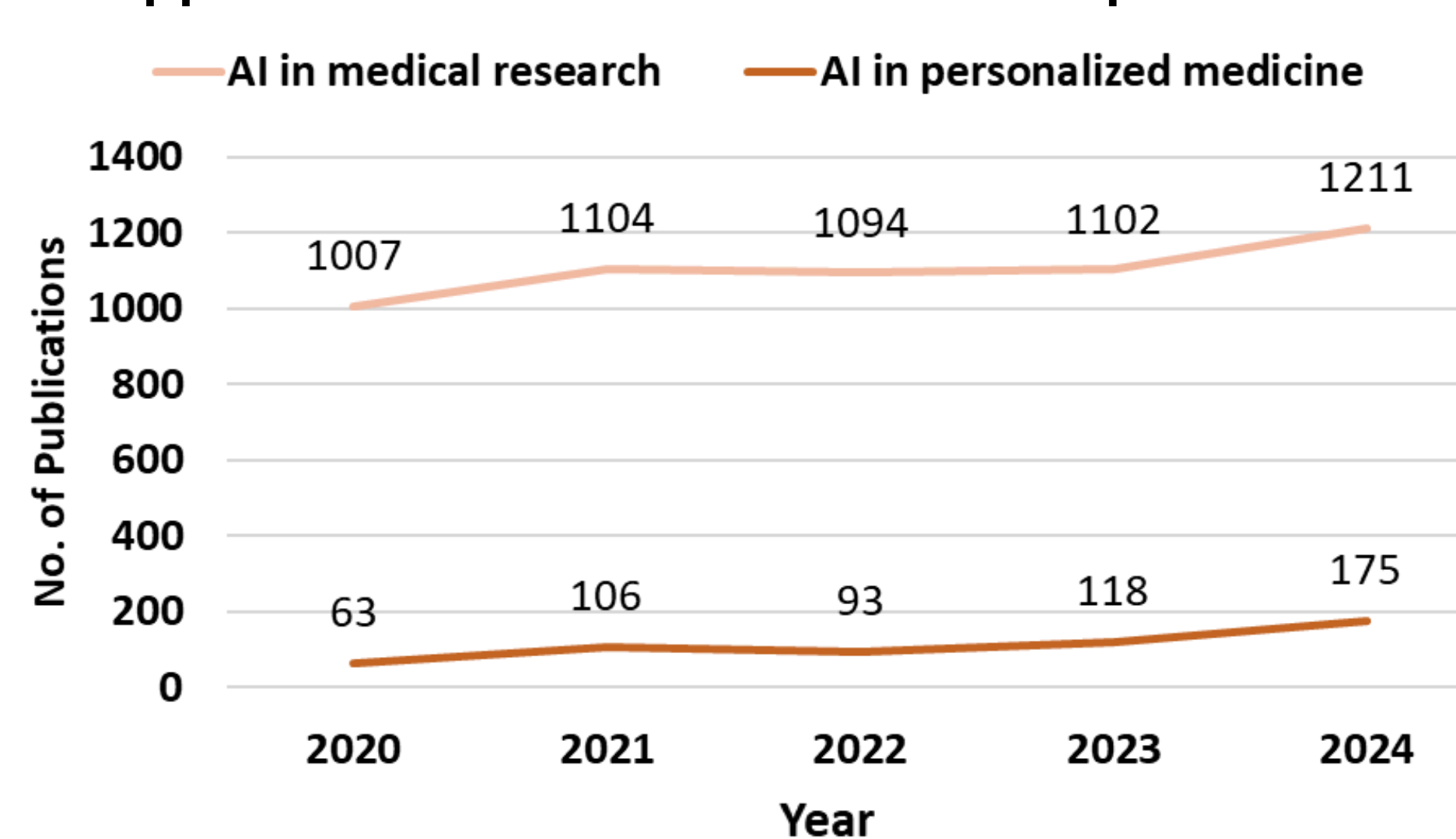


Fig. 1: Graph represents the number of articles published on PubMed in last 5 years about (a) using AI in Medical Research and (b) application of AI for personalized medicine

Quantum Computing in Drug Development

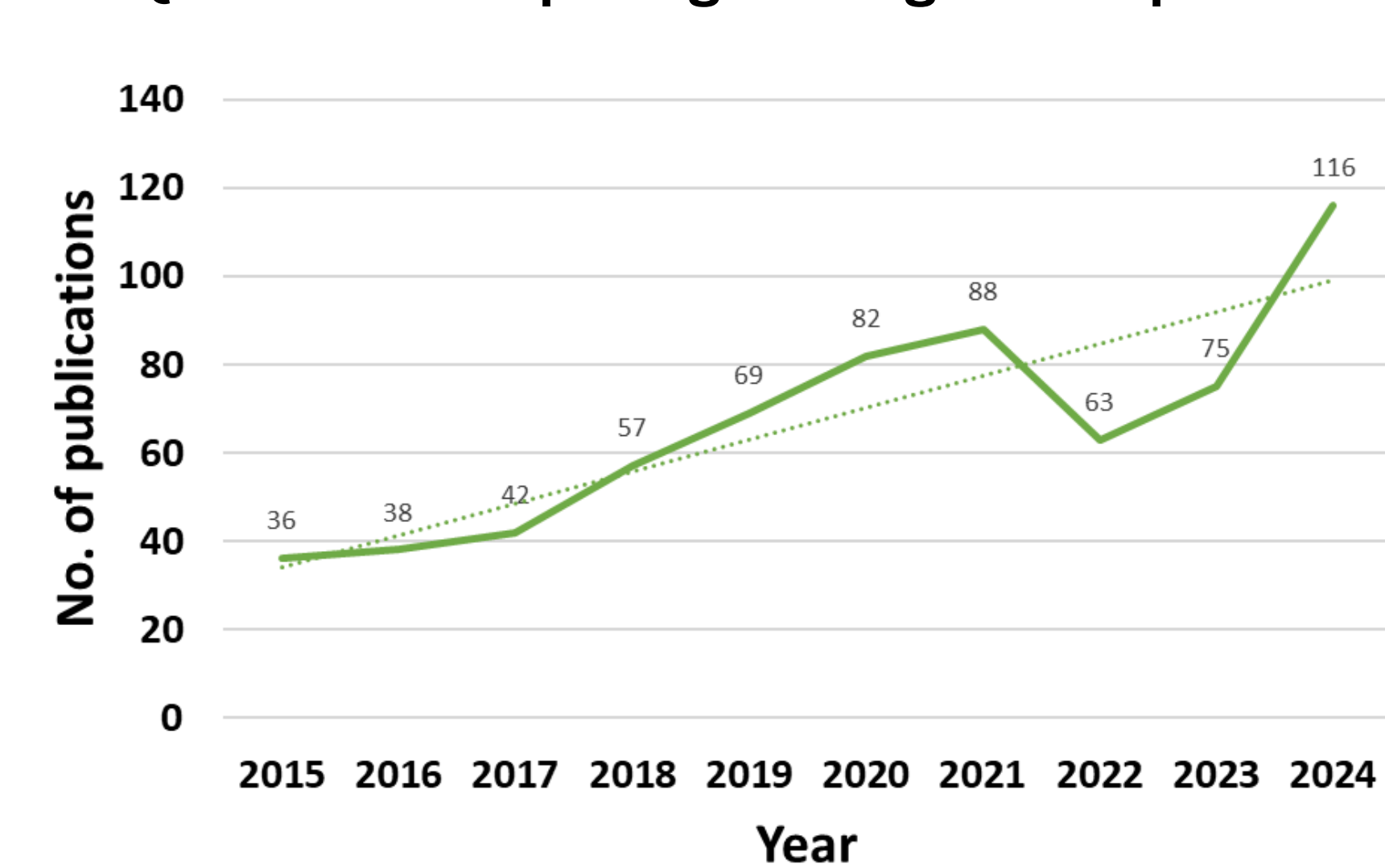


Fig. 2: The graph shows an up-trend in the research pertaining to apply QC in drug development (DD) based on number of articles published in last ten years.

Application of AI in various fields in past 5 year

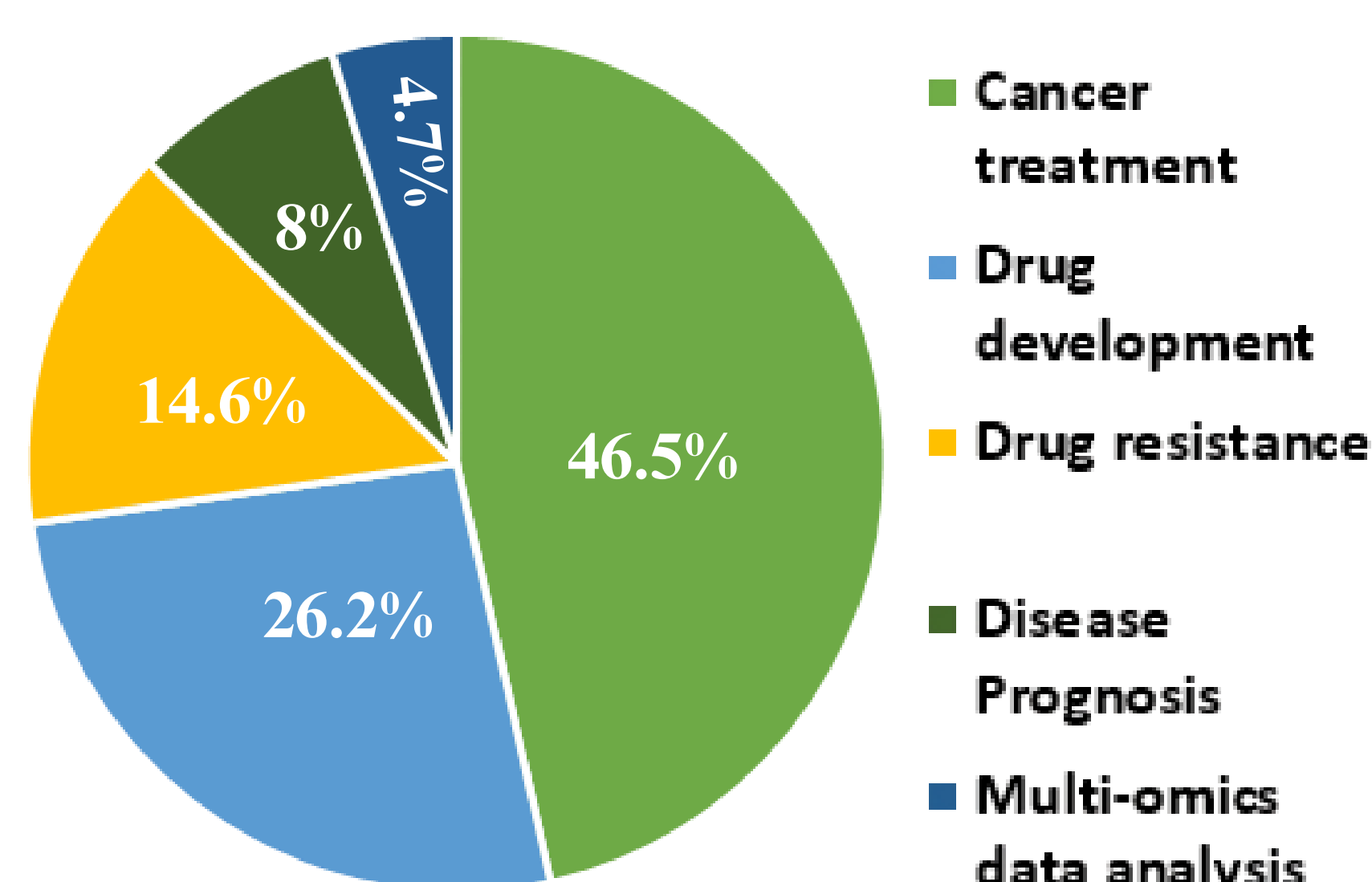


Fig. 3: Above chart represents the percentage distribution of use of AI in major subgroups of the analyzed publications. Mainly being in cancer treatment domain followed by DD.

Application of QC in Drug Development

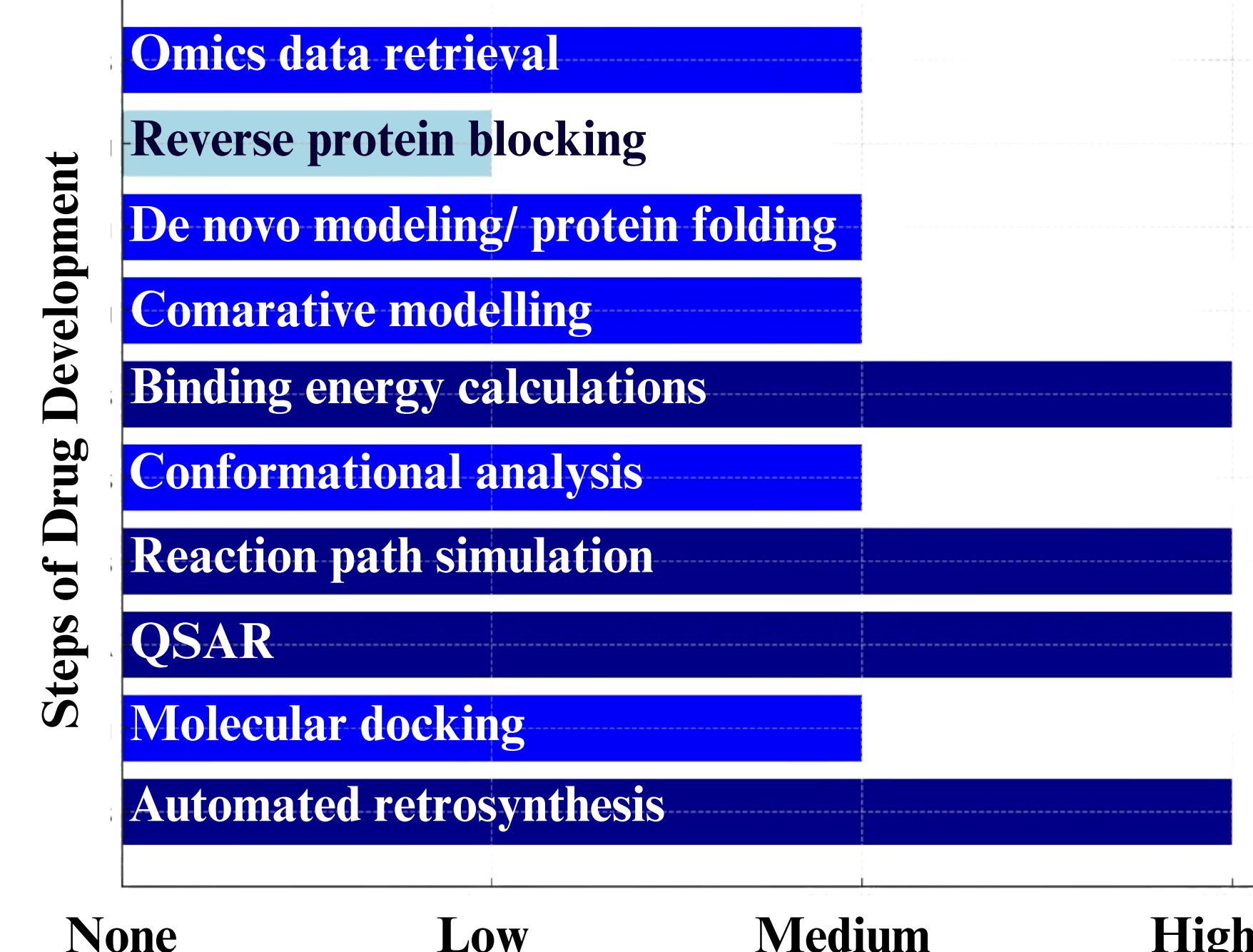


Fig. 4: Expected applicability of Quantum Computing in various steps of drug development ranging Low to High [5].

Discussion

- AI algorithms in biomedical research excel at identifying the hidden and subtle targets/biomarkers through genomic, proteomic, and clinical data and predicting the onset or progression of disease.
- Quantum computing and quantum machine learning provide unprecedented precision in simulating molecular interactions and predicting drug efficacy, drastically reducing time and cost for newer drug developments.
- AI can analyze individual genomes for different treatment responses and predict optimal drug dosage and treatment plans, reducing adverse effects and maximizing therapeutic effects.
- With future integration of AI and QC in biomedical research, it is crucial to address ethical considerations related to data privacy, algorithmic bias, and equitable access to these technologies.

References

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